

Date: 16/07/2025

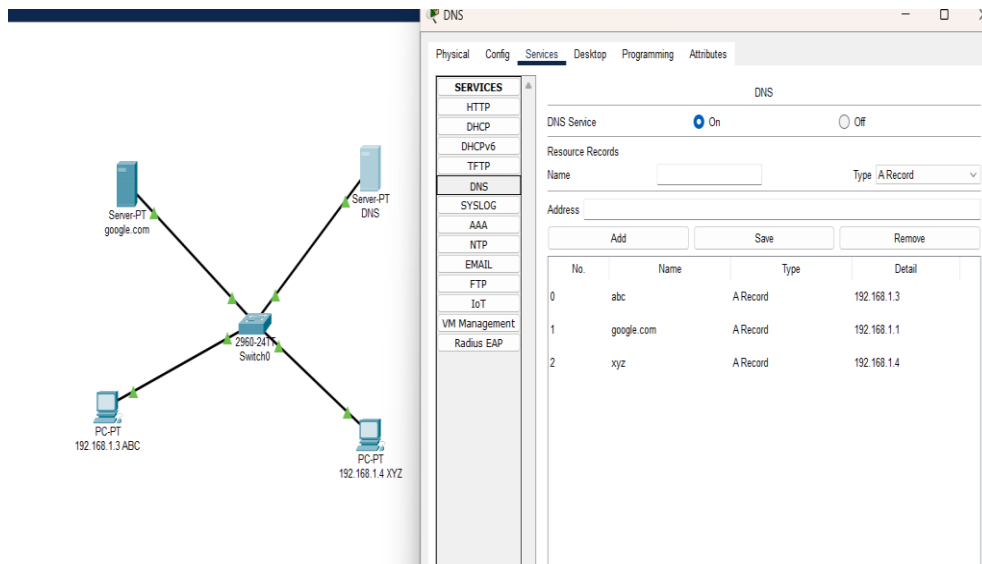
Lab Practical #07:

Study the application layer protocol DNS, DHCP, FTP.

Practical Assignment #07:

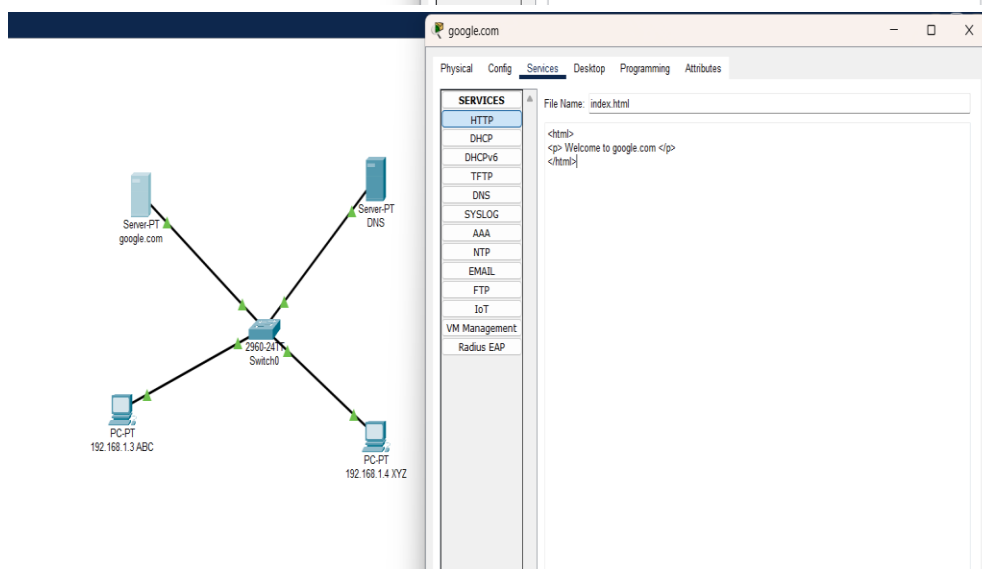
1. Implement the application layer protocol DNS, DHCP, and FTP. Also check connectivity between them using ping command or PDU utility.

- DNS:**



The image shows a network diagram on the left and a DNS configuration window on the right. The network diagram features a central switch labeled '2960-24T Switch0' connected to four devices: 'Server-PT google.com', 'Server-PT DNS', 'PC-PT 192.168.1.3 ABC', and 'PC-PT 192.168.1.4 XYZ'. The DNS configuration window has tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'SERVICES' tab is active, showing a list of services including HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, and Radius EAP. The 'DNS' service is selected, and the 'DNS Service' is set to 'On'. Below this, there is a 'Resource Records' section with a table listing three records:

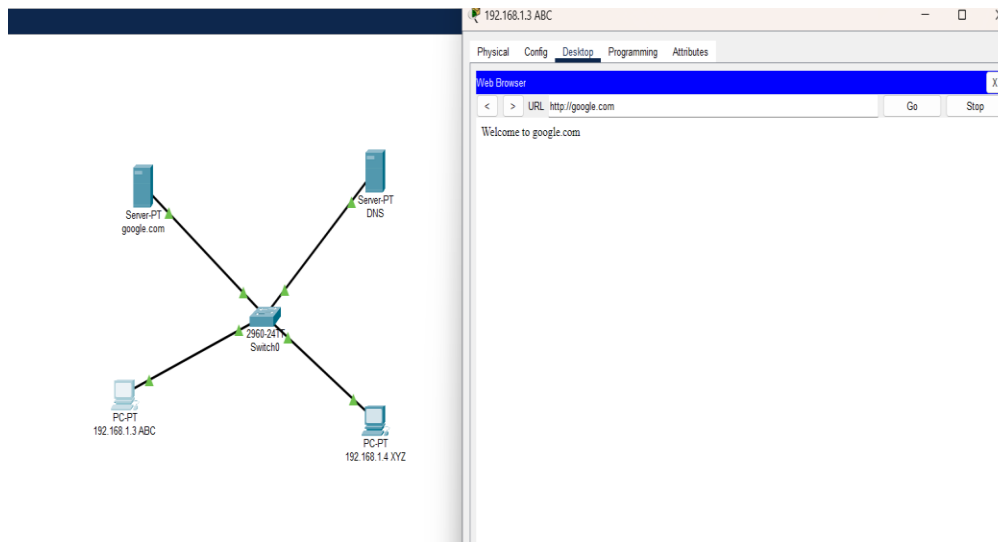
No.	Name	Type	Detail
0	abc	A Record	192.168.1.3
1	google.com	A Record	192.168.1.1
2	xyz	A Record	192.168.1.4



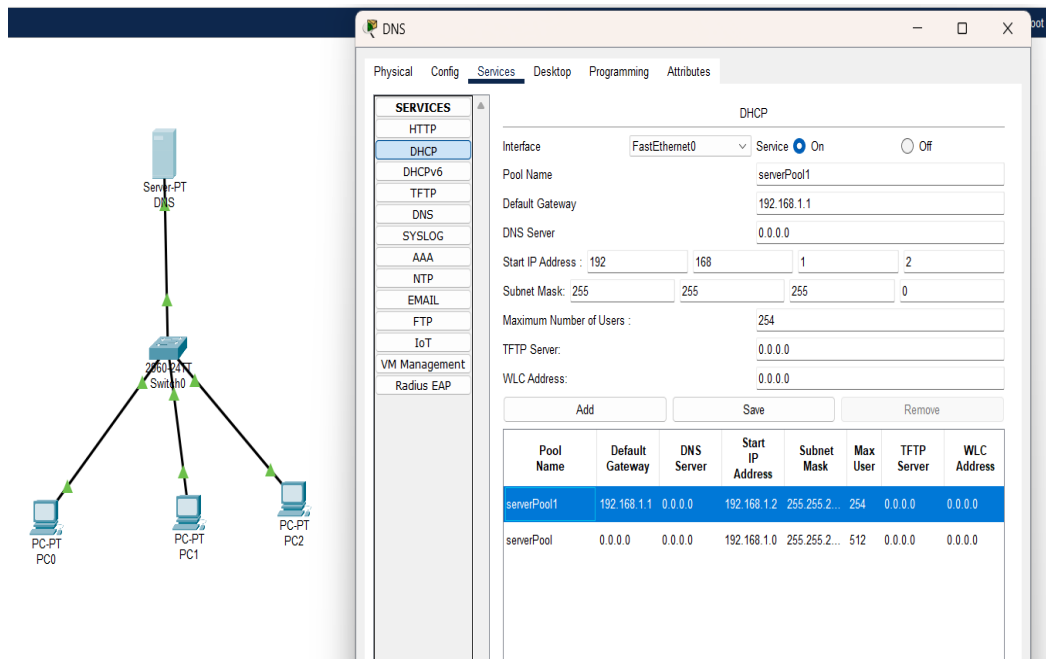
The image shows the same network diagram as above on the left and a web browser window on the right. The browser window displays the 'google.com' page. The 'SERVICES' tab is active, showing a list of services including HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, and Radius EAP. The 'DNS' service is selected, and the 'DNS Service' is set to 'On'. Below this, there is a 'Resource Records' section with a table listing three records:

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- **DHCP:**



DNS

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool1

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

Start IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Maximum Number of Users: 254

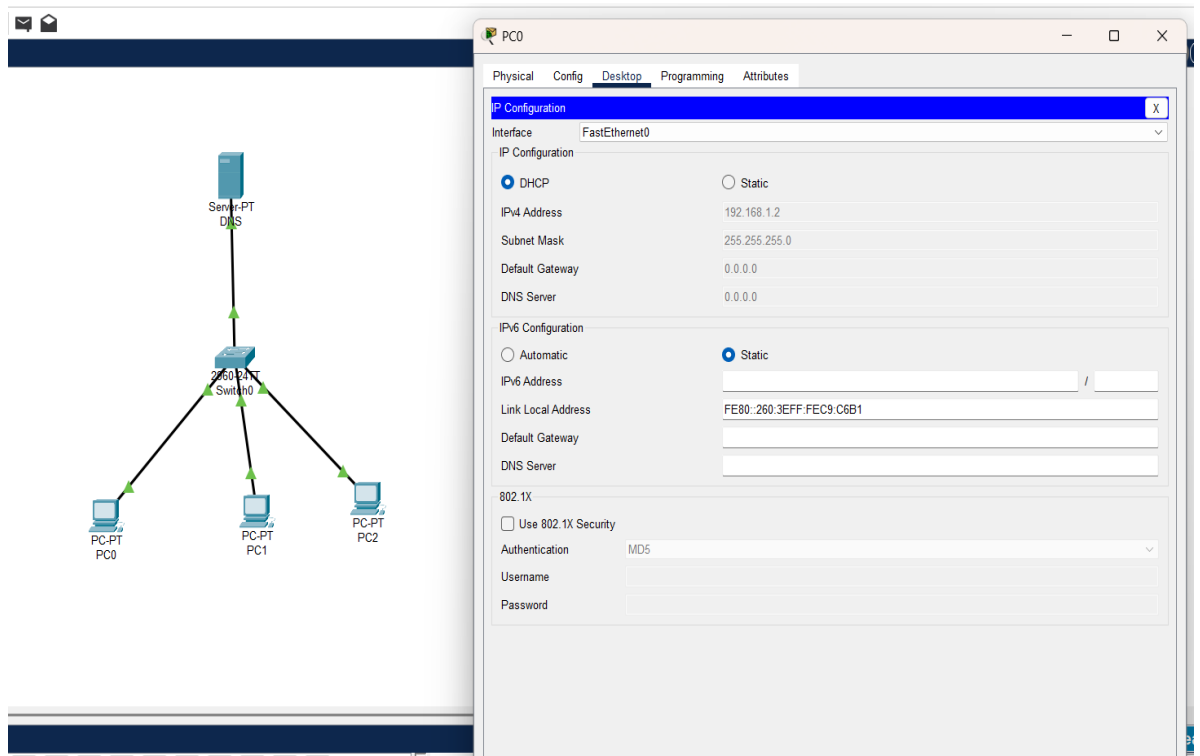
TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

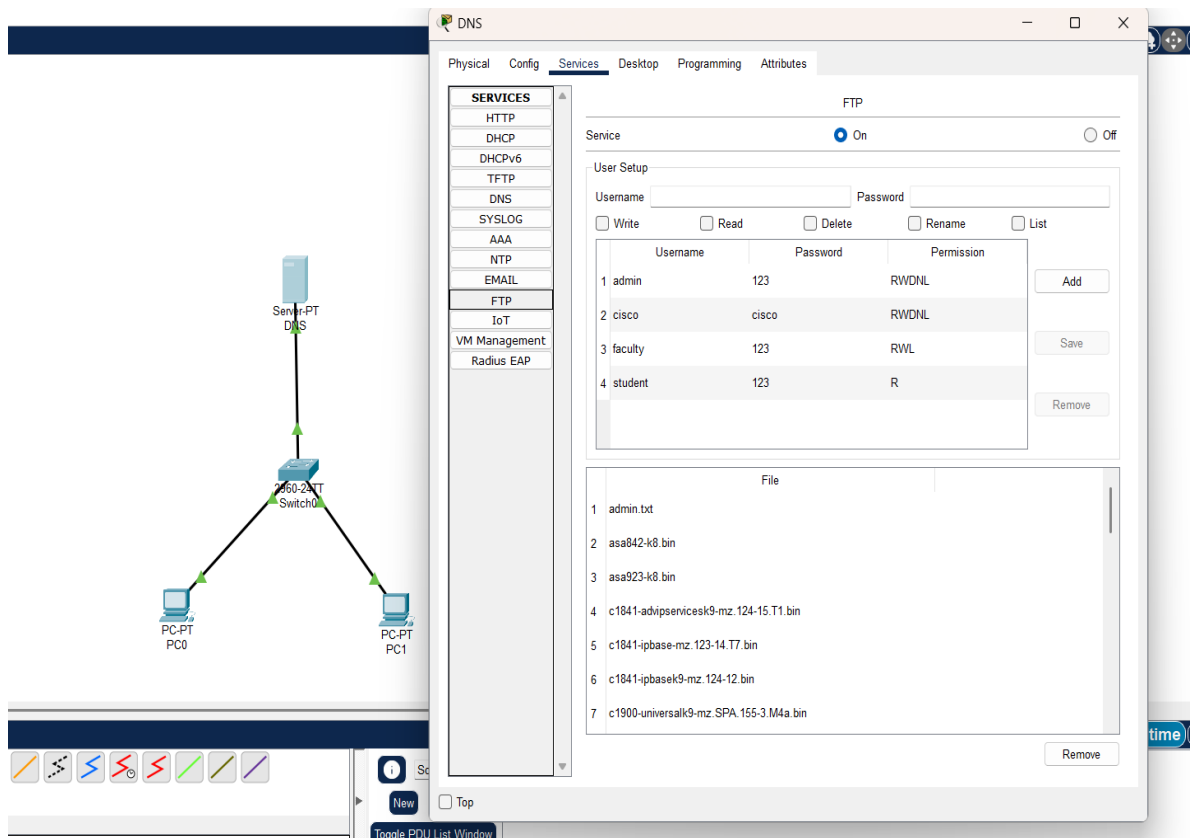
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool1	192.168.1.1	0.0.0.0	192.168.1.2	255.255.255.0	254	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.1.0	255.255.255.0	512	0.0.0.0	0.0.0.0

Date: 16/07/2025



The image shows a network diagram on the left and the configuration window for PC0 on the right. The network diagram illustrates a topology where a Server-PT (DNS) is connected to a 2600-24T Switch0, which is then connected to three PC-PTs (PC0, PC1, and PC2). The PC0 configuration window is open to the 'Desktop' tab, showing the 'IP Configuration' section. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'DHCP' is selected. The 'IPv4 Address' is 192.168.1.2, the 'Subnet Mask' is 255.255.255.0, the 'Default Gateway' is 0.0.0.0, and the 'DNS Server' is 0.0.0.0. The 'IPv6 Configuration' section shows 'Static' selected, with an 'IPv6 Address' field and a 'Link Local Address' of FE80::2603:EFF:FE09:C6B1. The '802.1X' section has 'Use 802.1X Security' unchecked, 'Authentication' set to MD5, and fields for 'Username' and 'Password'.

• FTP:

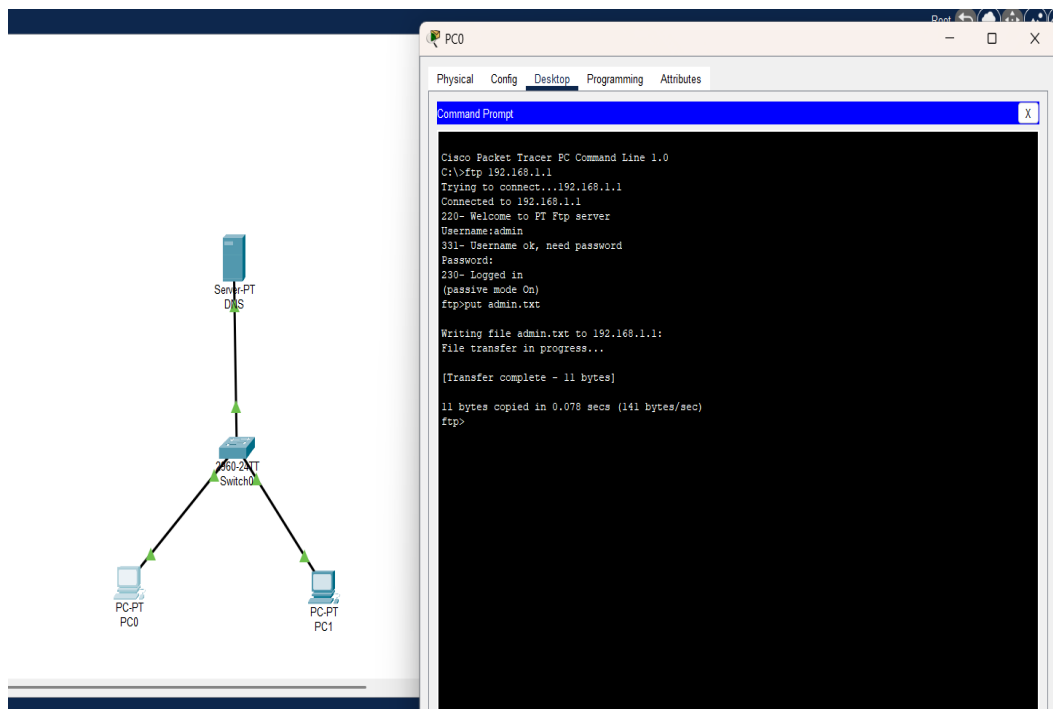
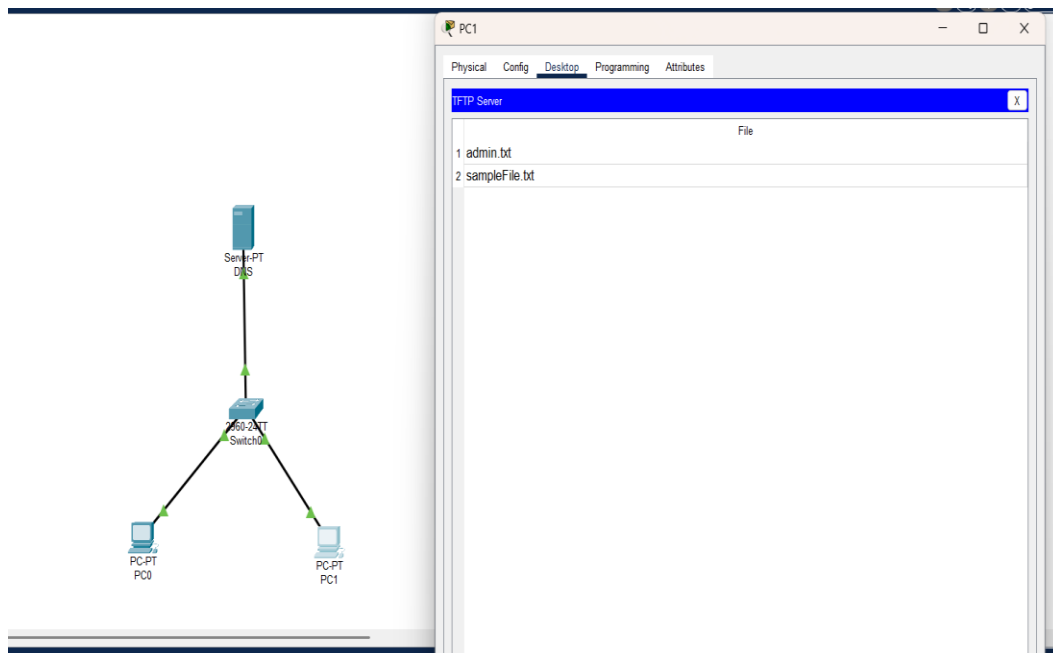


The image shows the same network diagram as above, but with the configuration window for the DNS server open to the 'Services' tab. The 'FTP' service is listed in the 'SERVICES' list on the left and is currently 'On'. The 'User Setup' section shows a table of users with their usernames, passwords, and permissions. The 'File' section shows a list of files and directories.

	Username	Password	Permission
1	admin	123	RWDNL
2	cisco	cisco	RWDNL
3	faculty	123	RWL
4	student	123	R

	File
1	admin.txt
2	asa842-k8.bin
3	asa923-k8.bin
4	c1841-advipservicesk9-mz.124-15.T1.bin
5	c1841-ipbase-mz.123-14.T7.bin
6	c1841-ipbasek9-mz.124-12.bin
7	c1900-universalk9-mz.SPA.155-3.M4a.bin

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